

Translation



The following Chinese industrial policy is designed to spur the development of brain-computer interface (BCI) technology, which allows humans to control or transmit information to computers or other devices directly with their brains. The policy anticipates and encourages applications of BCI technology in areas such as medical treatment, wearable displays, and surveillance in the coming years.

Title

Implementation Opinions of Seven Ministries on Promoting the Innovative Development of the Brain-Computer Interface Industry

七部门关于推动脑机接口产业创新发展的实施

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Source

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Implementation Opinions of the Ministry of Industry and Information Technology, the National Development and Reform Commission, the Ministry of Education, the National Health Commission, the State-owned Assets Supervision and Administration Commission of the State Council, the Chinese Academy of Sciences, and the National Medical Products Administration on Promoting the Innovative Development of the Brain-Computer Interface Industry

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To the industry and information technology, development and reform, education, health, state-owned assets, and medical products authorities of all provinces, autonomous regions, province-level municipalities, cities with independent planning status under the national economic and social development plan,¹ and the Xinjiang Production and Construction Corps; all subordinate entities of the Chinese Academy of Sciences (CAS); relevant central enterprises;² and all relevant entities:

Brain-computer interfaces (BCI) establish an information channel between the brain and machines to realize coordinated interaction between biological intelligence and machine intelligence. They are a cutting-edge technology arising from the integrated development of life sciences and information sciences. At present, innovative achievements in BCIs continue to emerge, the industry is rapidly growing, and disruptive breakthroughs are taking shape. BCIs have already become an important area for the in-depth integration of scientific and technological (S&T) innovation and industrial innovation. To seize the opportunities presented by a new round of S&T revolution and industrial transformation, promote high-quality development of the BCI industry, accelerate the formation of new quality productive forces (新质生产力), empower new-style industrialization (新型工业化) at a high level, and effectively support the building of a modernized industrial system, these *Implementation Opinions* are hereby formulated.

I. Overall Thinking

Guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, thoroughly implementing the spirit of the 20th Party Congress and the Second and Third Plenums of the 20th Central Committee, we will root ourselves in the new stage of development (新发展阶段), implement the new concept of development (新发展理念), promote the building of the new pattern of development (新发展格局), and do overall planning for development and security. With enhancing innovation capacity in the BCI industry as the main focus, driven by the integrated development of life sciences and information sciences, and guided by the expansion of application scenarios, we will take the development of high-performance complete-machine (整机)

¹ Translator's note: The five Chinese "cities with independent planning status under the national economic and social development plan" (国家社会与经济发展计划单列市; 计划单列市) are Dalian, Qingdao, Ningbo, Xiamen, and Shenzhen.

² Translator's note: The term "central enterprises" (中央企业; 央企) refers to state-owned enterprises (SOEs) directly under the Central People's Government of the People's Republic of China (aka the State Council) whose leaders are appointed by the State Council or the Chinese Communist Party (CCP) Central Committee, either directly or by the CCP Central Organization Department, the State-owned Assets Supervision and Administration Commission (SASAC), or another central Party or government department.

BCI products as the key lever, strengthen forward-looking planning and policy guidance, and accelerate the cultivation and formation of new tracks for future industries.

By 2027, we will have achieved breakthroughs in key BCI technologies, and an advanced technological, industrial, and standards system will have been preliminarily established. The performance of electrodes, chips, and complete machines will reach internationally advanced levels, and BCI products will see accelerated adoption in industrial manufacturing, medical and health services, and lifestyle and consumer goods. The industry will continue to expand in scale, establishing two to three industrial development cluster zones and developing a batch of new scenarios, new models, and new business formats (新业态).

By 2030, the innovation capacity of the BCI industry will be significantly enhanced, forming a safe, secure,³ and reliable industrial system. Two to three globally influential leading enterprises and a group of professional, precision, specialized, and innovative (专精特新) small- and medium-sized enterprises (SMEs) will be cultivated, building an internationally competitive industrial ecosystem and bringing China's comprehensive strength to the forefront of global rankings.

II. Strengthen Efforts in Basic Hardware and Software

- (1) Innovate brain-signal sensing components. We should develop implantable electrodes for different regions, including the periosteal, meningeal, and intracortical areas; explore cerebrovascular implantable electrodes; strengthen material stability and reliability; develop dedicated preparation and packaging processes; and improve electrode channel count, biocompatibility, spatial resolution, and signal-to-noise ratio. We should accelerate R&D of highly usable (高可用), self-adaptive non-implantable electrodes; develop new low-impedance, thin-medium (薄介质) electrode materials; and enhance electrode convenience, comfort, and ease of use. We should innovate new brain signal sensors based on optical, electrical, magnetic, ultrasonic, and chemical modalities, overcome the limitations of monomodal signals, and improve brain signal sensing capability.
- (2) Achieve breakthroughs in key BCI chips. We should develop high-channel-count, high-speed brain signal acquisition chips; strengthen

³ Translator's note: The Chinese word 安全 encompasses the meanings of both "safety" (protection from accidental harm) and "security" (protection from deliberate harm). In this translation, it is variously translated as "safety," "security," or "safety and security" at the translator's discretion.

analog-to-digital conversion, channel management, and noise suppression; and enhance the amplification capability for brain signal acquisition. We should develop high-performance, ultra-low-power brain signal processing chips; strengthen parallel processing capacity; and promote integrated sensing, computing, and modulation functions. We should develop ultra-low-power, high-speed, and highly reliable communication chips to improve brain signal transmission and anti-interference capability.

- (3) Consolidate the software tool foundation. We should improve brain signal encoding and decoding software, reduce the cognitive load of the encoding process, apply artificial intelligence (AI) technologies to strengthen decoding capacity and task iteration optimization capability, and enhance encoding–decoding accuracy, response speed, and scenario generalizability. We should develop dedicated control–interaction software, improve the precision of device control and neural regulation, and strengthen multi-task collaborative processing capability. We should build dedicated operating systems and general-purpose software platforms to achieve multimodal data integration, encoding–decoding algorithm compatibility, and configurable interaction software functions, and optimize the user experience.

Box 1: Core Software and Hardware Strengthening Projects

Highly integrated components. We should conduct research on highly integrated electronic packaging technologies, reduce the power consumption and noise of integrated chips, and create BCI chips that integrate signal perception, computing, and modulation. We should promote integration of implantable electrodes and signal-acquisition chips, develop high-density feedthrough devices, and advance the integration of electrodes, chips, and power supply units to meet the transmission requirements of ultra-flexible neural electrode channels. We should develop on-chip intelligent BCI interaction devices and explore new solutions based on hybrid intelligence (混合智能) and brain-inspired computing.

Paradigm encoding software. We should accelerate breakthroughs in integrated paradigm collaborative control technologies (混合范式协同控制技术), and develop dedicated encoding software for paradigms such as perception, movement, language, and emotion, improving psychological and cognitive assessment, sentiment analysis, and interventional therapeutic capability. We should design new natural and imperceptible interaction methods, build user-friendly encoding software, and reduce the influence of external stimuli.

Neural decoding software. We should carry out research on decoding algorithms based on AI technologies, achieve breakthroughs in weak neural signal acquisition, electroencephalogram (EEG) feature recognition, rapid decoding of BCI commands, decoding of subtle movements, auditory-visual decoding, and other technologies, and realize efficient and precise recognition of brain intentions (脑意图). We should conduct research on transfer learning methods for neural decoding, reduce dependence on sample data, and enhance the cross-device, cross-test-subject, and cross-task capabilities of decoding software.

III. Build High-Performance Products

- (4) Accelerate breakthroughs in implantable device R&D. We should explore new products that integrate high-density neural recording sensors and ultra-low-power implantable chips, innovate brain intention recognition functions, and improve control accuracy and response speed. We should improve mature products such as unidirectional and bidirectional deep brain stimulators, responsive electrical stimulators, and cochlear implants, enhance the accuracy of signal acquisition and functional modulation, and strengthen the efficacy of neurostimulation.
- (5) Promote mass production and iterative upgrades of non-implantable devices. We should innovate form factors such as forehead-mounted,

ear-mounted, in-ear, and hairclip-style devices, and push non-implantable products toward lightweight, high-speed, and low-power designs. We should develop integrated BCI products such as helmets, head-mounted displays, glasses, and headphones, and promote their integration with existing lifestyle and consumer products to support iterative application and large-scale adoption of non-implantable products.

- (6) Develop auxiliary devices. We should develop auxiliary physiological signal devices, and improve the accuracy of interaction control and perceptual evaluation capabilities through the fusion of brain signals with multimodal signals such as electromyography, electrooculography, electrocardiography, and near-infrared signals. We should do R&D of high-precision surgical robots for implantable BCIs, achieve breakthroughs in submicron-level precision control and dynamic adjustment technologies, and enhance capabilities for localized high-precision real-time imaging and three-dimensional reconstruction.

Box 2: Complete Machine Engineering Projects
Perceptual evaluation products. We should develop portable and wearable perceptual signal-collection devices and evaluation systems, integrate and process multimodal signals such as electromyography, electrooculography, and electrocardiography, conduct real-time analysis and provide visualized feedback data, and achieve efficient perception and evaluation of fatigue, insufficient attention, delayed reactions, and abnormal postures, helping users make corrective feedback adjustments. Emotion detection products. We should develop EEG-based emotion state detection systems, support emotion evaluation tasks using multimodal information such as images and audiovisual data, apply AI algorithms to improve the accuracy of emotion classification, achieve real-time emotion detection and evaluation, and improve closed-loop emotion regulation functions. Controlled interaction products. We should develop brain-controlled robots, brain-controlled computers, and brain-controlled home appliances, strengthen control accuracy, response speed, and interaction friendliness, and achieve capabilities for handling complex tasks. We should develop exoskeleton products based on EEG signal feedback, and strengthen the intuitiveness and precision of control modes. Security identification products. We should develop security identification systems that integrate EEG and other physiological feature signals, achieve breakthroughs in identity recognition technologies for wearable devices, explore the application of

brain-computer identity verification technologies in scenarios such as security authentication and payment transactions, and enhance security and privacy protection.

Neural regulation products. We should continue improving adaptive neural regulation products based on electromagnetic stimulation, and make forward-looking plans for products using optical and ultrasonic neurofeedback modulation methods, enable the treatment of neurological and psychiatric diseases, and explore applications in emotion regulation, holistic health (康养) and health maintenance, and cognitive enhancement (认知提升).

Perceptual reconstruction products. We should do R&D of multichannel visual and auditory cortex neural information detection and stimulation systems to achieve reconstruction and enhancement of auditory and visual perception.

IV. Promote the Application of Technological Achievements

- (7) Promote industrial innovation achievements. We should organize winner-takes-all open competitions⁴ for key tasks, support well-positioned organizations (优势单位) in hosting high-level competitions and exhibitions, and comprehensively identify innovative achievements. We should regularly select technologically advanced, demonstrably effective, and broadly replicable exemplary cases, give full play to their leading and driving role, and promote the large-scale application benefits of BCI technologies. We should organize industry supply-demand linking (供需对接) activities, build platforms for industry empowerment and linking, remove barriers in industry applications, and accelerate the implementation of innovative achievements.
- (8) Improve testing and pilot-scale capabilities. We should support the development of testing and evaluation institutions, establish product testing specifications, develop dedicated instruments for brain signal inspection and testing, improve supporting assessment tools, build testing environments for BCI devices and systems, strengthen the construction of verification platforms, and enhance product application testing capability. We should plan and build pilot-scale platforms for products, accelerate the R&D and commercialized application of new products, strengthen capabilities for signal acquisition and processing and algorithm verification,

⁴ Translator's note: The idea behind "winner-takes-all open competition" (揭榜挂帅), in the context of Chinese science and technology projects, is that the government openly lists the technological breakthrough(s) it desires. Any individual or group in society, not just a select few, are then eligible to win a cash award if they succeed in making the breakthrough.

improve product interoperability and ease of use, and enhance product quality.

Box 3: Application Expansion Projects

Discover industrial manufacturing applications. We should guide applications in key industrial links such as R&D design, production and manufacturing, and operations and maintenance management; build an industrial metaverse integrating life and machinery (构建生命机械一体化的工业元宇宙); and innovate industrial equipment and robot-control solutions that rely on BCIs. We should promote early pilot use in key industries such as hazardous materials, nuclear energy, mining, and electric power; build industrial safety management platforms; provide early warning for events such as hypoxia, poisoning, and fainting; and ensure production and personnel safety. Facing the needs of special industrial scenarios such as deep sea, deep space, high plateaus, and extremely cold regions, we should reduce equipment operation thresholds, establish “one-to-many” and “many-to-many” human-computer interaction models, and improve operational efficiency in complex scenarios.

Accelerate medical and health applications. Facing clinical diagnosis and treatment scenarios involving neurodegenerative diseases and psychiatric disorders, we should explore pathological and physiological mechanisms, innovate neuromodulation technologies, and effectively improve disease diagnosis and treatment outcomes. Facing the rehabilitation needs of patients with motor, speech, or sensory impairments, we should develop new solutions for movement assistance, communication assistance, and perception recovery, and improve patient rehabilitation quality. Facing brain disease prevention and health management scenarios, we should use BCI products to monitor and analyze brain activity characteristics in real time, prevent potential health problems, and reduce the risk of severe illness among patients.

Explore lifestyle and consumer applications. We should focus on new digital consumption scenarios and strengthen capabilities for remote control, virtual perception, and immersive interaction. Focusing on traffic safety needs, we should use BCIs to monitor drivers’ physiological states, provide timely reminders for abnormal conditions such as fatigue, insufficient attention, and delayed reactions, and reduce the probability of traffic accidents. Focusing on the needs of sports training and brain intelligence cultivation (脑智培育), we should optimize multi-vital-sign health monitoring and management solutions, and, relying on brain and neural efficiency evaluation systems, build personalized training and competition systems and improve training outcomes and athletic performance.

V. Strengthen Innovation Entities

(9) Cultivate well-positioned enterprises (优势企业). We should develop and

expand leading enterprises in the BCI field, support the establishment of industrial innovation consortia, and have them take the lead in undertaking national S&T major special projects (国家科技重大专项) and other tasks. We should promote innovation supply by BCI SMEs and accelerate the cultivation of S&T- and innovation-oriented SMEs, high-tech enterprises, professional, precision, specialized, and innovative SMEs, professional, precision, specialized, and innovative “little giant” enterprises, manufacturing single-item champion (单项冠军) enterprises, as well as gazelle enterprises and unicorn enterprises. We should implement the Qihang Enterprise⁵ cultivation project and improve the tiered enterprise cultivation system (企业梯度培育体系). We should promote integrated innovation among small, medium-sized, and large enterprises, and encourage leading enterprises to open algorithm frameworks, public data, and general purpose technologies to the public, creating a favorable environment for enterprises to grow together.

- (10) Improve innovation vehicles. We should support the establishment of national manufacturing industry innovation centers, national technology innovation centers, national emerging industry innovation centers (国家新兴产业创新中心), national industrial technology commercialization centers (国家产业技术工程化中心), national key laboratories (全国重点实验室), and national clinical medicine research centers (国家临床医学研究中心), bring together the strengths of all parties from industry, academia, and research institutes, and enhance the supply capacity for key and general purpose BCI technologies. Relying on organizations such as industry alliances and industry associations, we should promote BCI technology exchange, supply-demand linking, and international cooperation, and deepen the integration of innovation chains, production chains, funding chains, and talent chains. We should build open-source communities, strengthen support for key open-source projects, and gather global developers for collaborative innovation.
- (11) Promote industrial clustering. We should guide innovation factors of production (创新要素) to concentrate in regions with strong foundations and high potential, build incubators and industrial parks in a graded and categorized (分级分类) manner based on local characteristics and industrial

⁵ Translator's note: China's State-owned Enterprise Supervision and Administration Commission (SASAC) in 2023 launched a campaign to designate promising startup state-owned enterprises (SOEs) in strategic emerging industries such as AI, quantum, and biotech as "Qihang Enterprises" (启航企业) and to nurture them with preferential policies, extra resources, and additional funding.

advantages, develop well-positioned clusters with strong innovation capability and high-quality application scenarios, and promote clustered development across the upstream and downstream of production chains. We should promote cooperation between BCI enterprises and enterprises in fields such as AI, new materials, and robotics, carry out joint technological and application research, and build an innovative environment for cross-domain collaboration.

VI. Enhance Industrial Support Capabilities

- (12) Strengthen standards leadership. We should establish a BCI technology standards system and lay out a roadmap for standardized development. All departments, in accordance with their respective responsibilities, should organize the formulation of technical standards, product standards, service standards, and testing standards, and accelerate the development of key standards based on industry development needs. We should carry out in-depth propaganda and promotion on standards to advance the implementation of standards. We should actively participate in international standards development and vigorously push Chinese standards to "go global" ("走出去").
- (13) Improve safety and security assurance. We should continuously advance ethics research, establish a sound governance system featuring departmental coordination and participation from society, and promote the coordinated development of both technological innovation and S&T ethics. We should establish a data governance framework, regulate the collection, storage, and use of user information, prevent brain privacy leaks (脑隐私泄露), and enhance our protection capabilities for biological digital information security.
- (14) Optimize talent development. We should strengthen talent development in relevant academic disciplines and majors, and plan and build specialized colleges⁶ such as colleges of future technology (未来技术学院) and colleges of modernized industry (现代化产业学院). We should encourage enterprises to cooperate with universities, scientific research institutes, and medical institutions to jointly cultivate interdisciplinary,

⁶ Translator's note: "Specialized colleges" (特色学院) are schools set up within Chinese universities to train students in technologies essential to strategic Chinese industries. These schools usually involve the government, companies, or research institutes as well as university personnel in funding, course design, and instruction. In theory, they should all be set up in universities or localities with special strengths in the given industry or technology, hence their "specialized" nature.

hybrid (复合型), engineering-oriented (工程型) talent, and enhance the supply of high-level talent. We should build practice bases for outstanding engineers, strengthen vocational education and technical training, and vigorously cultivate skilled talent oriented toward industrial applications. We should strengthen overseas talent exchange and recruitment, improve the talent service system, and ensure that talent can be both attracted and retained.

VII. Assurance Measures

- (15) Strengthen overall coordination. Under the leadership of the Central Science and Technology Commission, we should strengthen interdepartmental coordination and advance work related to technological breakthroughs, industrial development, industry applications, and safety and security governance in an integrated manner. We should deepen central-local cooperation, optimize industrial layouts, encourage localities to formulate targeted and workable policy measures based on their actual conditions, and promote technological innovation and industrial development in accordance with local circumstances.
- (16) Strengthen policy support. We should strengthen technological breakthroughs and application demonstrations of BCI, promote the planning and implementation of a number of major projects, and strengthen collaborative breakthrough capabilities. We should encourage increased investment by the National Manufacturing Industry Transformation and Upgrading Fund (国家制造业转型升级基金), the China SME Development Fund (国家中小企业发展基金), and other funds; implement the "S&T Industry-Finance Integration" Special Projects (“科技产业金融一体化”专项); give full play to the role of the National Industry-Finance Cooperation Platform; and drive greater capital investment. We should strengthen registration guidance and provide priority support for key products such as implantable BCI medical devices. We should make full use of insurance and compensation policies for first units, sets, and batches of equipment and accelerate commercialization.
- (17) Deepen international cooperation. We should pool domestic and international innovation resources, strengthen technological R&D and industrial exchange, encourage foreign enterprises and institutions to establish R&D centers, manufacturing bases, and other facilities in China, and accelerate the internationalized development of the industry. We should support the expansion of overseas markets, promote outstanding

products and solutions, and develop an internationalized industrial development model. We should support Chinese enterprises and public institutions⁷ in participating in international conferences and actively contribute Chinese products, Chinese solutions, and Chinese wisdom.

Ministry of Industry and Information Technology

National Development and Reform Commission

Ministry of Education

National Health Commission

State-owned Assets Supervision and Administration Commission of the State Council

Chinese Academy of Sciences

National Medical Products Administration

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⁷ Translator's note: "Public institutions" (事业单位) are organizations created and led by Chinese government departments that perform a variety of functions. Notably, universities, research institutes, and hospitals in China tend to be public institutions. Unlike state-owned enterprises (SOEs), public institutions do not create material products and are non-profit. Public institutions are not considered government agencies, and their employees are not civil servants. Most public institutions are fully or partially government-funded, but some fully privately funded (but still government-led) public institutions exist.